

LAGGERI SHAILAJA

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Professional Summary

- System Engineer with **4.2 years** of experience in the MBSE development in **Automotive domain**, currently working in **TCS (Tata Consultancy Services)**.
- Skilled in **SysML** and **UML** modeling using **Cameo Systems Modeler (MagicDraw)** and **IBM Rational Rhapsody** for system architecture, behavior modeling.
- Hands-on experience in **SysMLv1** for manual model creation and **SysMLv2** for automated model generation using prompts, templates, and reusable components.
- Proficient in developing **Requirement, Use Case, Block Definition (BDD), Internal Block (IBD), Activity, Sequence, State Machine, Package, and Parametric Diagrams**.
- Worked on **ADAS** features such as **Automatic Emergency Braking (AEB)**, **Lane Keep Assist (LKA)** for automotive applications.
- Developed Engine system architecture models for GE Aerospace, synchronizing Cameo models with Siemens Teamcenter using the Synidea plugin for **MBSE-PLM integration**.
- Implemented **EV charging control models (AC/DC Enable–Disable)** for Stellantis using **SysMLv2** automation in IBM Rhapsody, improving model efficiency and reusability.
- Utilized **Jupyter Notebook** for Creation of SysML Text Notation errors and Model generation.
- Underwent comprehensive training in MBSE and SysML during the Ford Climate Offshore project, developing conceptual models (**Stopwatch, Temperature Sensor**) to strengthen modeling fundamentals.
- Strong collaborator with experience working directly with clients, architects, and domain experts to **review, refine, and validate models**.
- Ensures **traceability, compliance, and alignment** of system models with stakeholder and project requirements across multiple domains.
- Knowledge in **AUTOSAR, ASPICE, CAN, and Functional Safety (ISO 26262 – FUSA)** principles.

Skills

- Cameo Modeler Tool (MagicDraw)
- IBM Rational Rhapsody
- SysMLv1 / SysMLv2
- Teamcenter – syndiea plugin
- Jupyter Notebook
- ASPICE & AUTOSAR awareness
- Model Development
- Awareness on FUSA
- Model automation using SysMLv2 prompts/Templates
- Requirement Diagram
- Use case Diagram
- Block Definition Diagram (BDD)
- Internal Block Diagram (IBD)
- Activity Diagram
- Sequence Diagram
- State Machine Diagram
- Package & parametric Diagram
- UI Simulation in cameo

Work History

PROJECT: 3 – Stellantis Development, 05/2025 to Present Client: Stellantis

- Worked on **Lane Keep Assist (LKA)** using SysMLv1 model creation with UI simulation.
- Utilized IBM Rational Rhapsody Tool for model creation and automation of EV charging functionalities such as **AC/DC Charging Enable–Disable operations**.
- Modeled extensively using **SysMLv2 for prompt template creation, code generation**, and model automation to improve modeling accuracy and efficiency.
- Created **Sequence Diagrams** to visualize communication flows between components during charging enable/disable scenarios.
- Designed system structures and architectures for EV systems, representing charging topologies and architectural hierarchies using **Internal Block Diagrams (IBD)**.
- Developed **State Machine Diagrams** to represent system workflows and behavior across multiple charging operations (e.g., enable, disable, and fault conditions).
- Collaborated with cross-functional engineering teams to review, refine, and validate system models for completeness and compliance with **MBSE** and **Stellantis modeling standards**.
- Contributed to developing MBSE best practices for next-generation **EV system** modeling and integration.
- Utilized **Jupyter Notebook** for Creation of SysML Text Notation errors and Model generation.

PROJECT: 2 – GE Aerospace, 06/2023 to 02/2025

Client: General Electric (GE)

- Utilized **Cameo Modeler Tool** for development of SysML models supporting engineering system architecture and requirement models.
- Collaborated directly with the client team to interpret requirements, clarify **system behaviors, and validate model completeness and compliance** with GE modeling standards.
- Designed and maintained **Requirement Diagrams, Block Definition Diagrams (BDD), and Internal Block Diagrams (IBD)** to represent key system components and their interactions.
- Worked on **Requirement diagrams** given by customer and **Engine model** generation using SysML diagrams.
- Created **Activity Diagrams** and **State Machine Diagrams** to define system workflows and behavior logic across multiple subsystems.
- Developed **Sequence Diagrams** to visualize communication flows between system elements.
- Integrated Cameo models with **PLM (Teamcenter)** using the **Syndeia plugin** to enable synchronization of MBSE data and improved collaboration between design and systems engineering teams.
- Documented detailed procedures for using **Cameo** with **Teamcenter** via **Syndeia**, providing guidance for seamless data integration and cross-tool traceability.
- Worked with cross-functional stakeholders to ensure system models aligned with **engineering requirements, validation standards, and MBSE best practices**.
- Contributed to establishing MBSE implementation guidelines for enhanced collaboration between modeling, systems, and **PLM integration teams**.

PROJECT: 1-FORD CLIMATE OFFSHORE, 05/2022 to 04/2023

Client: Ford

- Underwent extensive training on **Cameo Modeler Tool** and **SysML** for Model-Based Systems Engineering (MBSE).
- Gained hands-on experience creating SysML diagrams such as **Use Case Diagrams, Block Definition Diagrams (BDD), Internal Block Diagrams (IBD), Activity Diagrams, and Sequence/State Machine Diagrams**.
- Modeled system behaviors and logical architecture elements for system **requirement traceability, validation, and functional decomposition**.

- Worked on system models representing **Stopwatch and Temperature Sensor**, strengthening system understanding and modeling proficiency.
- Worked on generation of **Automatic Emergency Braking (AEB)** system model using SysML diagram
- Collaborated with system engineers, architects, and domain experts for **model validation, architecture design, and model documentation** aligned with best MBSE practices.
- Ensured **traceability, completeness, and compliance** of system requirements within SysMLbased modeling frameworks.
- Developed a solid understanding of system architecture design principles, behavioral modeling, and implementation of best practices using **MBSE methodologies**.

Previous Work Experience

Worked as senior Associate in Wipro Limited from February 24, 2020 to May 12, 2021 (Non-IT)

Education

Masters of Technology: Embedded Systems, 09/2021

B V Raju Institute of Engineering - Medak

- Awarded Gold Medal in masters for academic excellence and securing top rank in the department.

Bachelor of Engineering: Electronics and Communication Engineering, 07/2019

DVR College of Engineering - Sangareddy

Disclaimer

I hereby declared that all the above-mentioned information is true to the best of my knowledge and belief.

(LAGGERI SHAILAJA)